

Supporting Information

Which modelling choices have been tested where marine survey data are sparse? An evidence map of 95 studies and a graded practitioner guide

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Appendix S1. Search strategy, screening, inclusion criteria, and non marine sources retained

S1.1 Databases and coverage

Searches combined method terms (Random Forest, boosted regression trees, VAST, sdmTMB, generalised additive model, MaxEnt, joint species distribution model, graph neural network), spatial terms (spatial autocorrelation, spatial cross validation, geostatistical, random field), data limitation terms (data limited, small sample, sparse) and marine ecosystem descriptors (marine, ocean, estuarine, fish, cetacean, benthic). Complete Boolean strings are in Table S2.

Table S1. Records identified, screened and included by source. Counts are unique records after duplicate removal within each source at each stage. Thirteen studies eligible at full text were duplicates across sources and were resolved at the final stage, so the per source included counts sum to 108 before that resolution and 95 after it.

Source	Period	Identified	After title and abstract	After full text	Included
Web of Science	2001 to 2025	185	89	58	50
Google Scholar	2001 to 2025	95	48	28	25
Discipline specific journals	2010 to 2025	35	25	14	13
Reference tracking		15	10	8	8
Total		330	172	108	95

Table S2. Complete Boolean search strings by topic area. Strings were executed across Web of Science and Google Scholar; the discipline specific journal search used the same topic terms applied to journal level search interfaces.

#	Topic area	Boolean search string	Rationale
1	General spatial ML and SDM	("spatial machine learning" OR "species distribution model*" OR "habitat model*" OR "niche model*" OR "SDM" OR "HSM") AND (marine OR ocean* OR aquatic OR fish* OR cetacean* OR "marine mammal*" OR benthos OR benthic OR pelagic OR demersal OR "marine ecosystem*")	Broad capture of spatial modelling approaches in marine systems
2	Random Forest	("random forest*" OR "RF" OR "ensemble learning" OR "random tree*") AND (marine OR ocean* OR fish* OR cetacean* OR aquatic) AND (spatial OR distribution OR habitat OR predict*)	Specific RF method with marine applications
3	Boosted Regression Trees	("boosted regression tree*" OR "BRT" OR "gradient boosting" OR "GBM" OR "boosted tree*" OR "gradient boost*") AND (marine OR ocean* OR fish* OR "marine species" OR cetacean*) AND (distribution OR habitat OR spatial)	BRT and gradient boosting variants
4	VAST framework	("VAST" OR "vector autoregressive spatio-temporal" OR "vector autoregressive spatiotemporal" OR "geostatistical delta" OR "spatio-temporal GLMM") AND (fisheries OR marine OR ocean* OR fish OR "stock assessment")	Purpose built marine spatio temporal framework
5	Spatial cross validation	("spatial cross-validation" OR "spatial cross validation" OR "spatial CV" OR "spatial autocorrelation" OR "spatial validation" OR "spatial blocking") AND ("species distribution" OR "habitat model*" OR SDM OR "predictive model*" OR "machine learning")	Critical validation methodology
6	GAM spatial models	("generalised additive model*" OR "GAM" OR "additive model*") AND spatial AND (marine OR ocean* OR fish* OR aquatic) AND (distribution OR habitat)	Traditional and hybrid spatial modelling approach
7	MaxEnt	("MaxEnt" OR "maximum entropy" OR "Maxent") AND (marine OR ocean* OR aquatic OR fish*) AND (species OR distribution OR habitat OR SDM)	Presence only modelling dominant method
8	Transfer learning	("transfer learning" OR "extrapolation" OR "transferability" OR "model transfer*" OR "spatial transfer*" OR "temporal transfer*") AND (species OR marine OR ecology OR distribution OR SDM) AND (predict* OR model*)	Transferability and forecast skill assessment
9	Data limited contexts	("data limited" OR "data-limited" OR "data poor" OR "limited sample*" OR "small sample*" OR "sparse data") AND (marine OR fisheries OR ocean*) AND (model* OR predict* OR distribution OR SDM)	Explicit focus on data constraints

Table S2 (continued)

#	Topic area	Boolean search string	Rationale
10	Graph neural networks	(“graph neural network*” OR “GNN” OR “deep learning” OR “neural network*” OR “convolutional neural”) AND (marine OR ocean* OR species OR ecology OR spatial) AND (distribution OR habitat OR predict*)	Emerging deep learning methods
11	Uncertainty quantification	(“uncertainty quantification” OR “prediction uncertainty” OR “model uncertainty” OR “adaptive monitoring” OR “value of information”) AND (“species distribution” OR marine OR ecology OR SDM) AND (spatial OR predict*)	Uncertainty and adaptive approaches
12	Spatial Random Forest specifically	(“spatial random forest” OR “spatial RF” OR “RFsp” OR “geographically weighted random forest”) AND (ecology OR species OR distribution OR marine OR spatial)	Enhanced spatial RF implementations
13	Ensemble methods	(ensemble OR “model averaging” OR “multi-model” OR “stacked model”) AND (“species distribution” OR SDM OR marine OR ecology) AND (predict* OR spatial)	Multiple model combinations
14	Validation reviews	(“cross-validation” OR “model validation” OR “performance assessment”) AND spatial AND (autocorrelation OR blocking OR structure) AND (ecology OR species OR SDM)	Methodological validation literature
15	Joint species distribution models	(“joint species distribution” OR “JSDM” OR “multivariate species” OR “HMSC” OR “hierarchical modelling of species communities” OR “community-level model”) AND (marine OR ocean* OR fish* OR aquatic) AND (spatial OR distribution OR predict*)	Joint modelling approaches leveraging data sharing

S1.2 Inclusion and exclusion criteria

Included: marine or estuarine application; a spatial modelling component; sample size reported; performance metrics or documented validation practice; peer reviewed and in English.

Excluded: purely terrestrial systems, except where a study provided a methodological result with no marine equivalent, in which case it was retained and flagged; reviews without original analysis; insufficient methodological detail to determine the spatial treatment or validation design; no performance assessment of any kind.

Corpus membership and citation are separate. The corpus comprises studies that passed the criteria above and contributed evidence to the synthesis. The reference list additionally contains statistical, algorithmic and software sources that have no study system at all, and which are cited for provenance or implementation rather than as evidence; these were never screened as corpus candidates and are not counted in any total reported in this review. Table S3 lists every non marine source cited, distinguishing those that anchor a finding from those cited for provenance only.

Table S3. Sources applied outside marine and estuarine systems, or to simulated data, cited in the main text. Each is identified as non marine at every point of use, and the consequence for the evidence base is addressed in main text Sections 3.1 and 4.4. The classification basis records how the system was determined: verified against the publisher record, from the article title, or from the journal scope.

Study	System	Reason for retention	Finding supported	Basis
<i>Studies anchoring a finding (corpus, retained under the methodological exception)</i>				
Hengl et al. (2018)	Soil and environmental benchmarks, terrestrial	The only study in the review that crosses spatial treatment against algorithm family on identical data; its kriging comparison also prevents the spatial structure finding being read as an endorsement of tree ensembles	Spatial structure finding (main text 3.2); Table 5	Title
Talebi et al. (2022)	Geoscience	Independent corroboration of the same contrast in a second domain, reported on error rather than discrimination metrics	Spatial structure finding (3.2)	Journal
Mushagalusa et al. (2024)	Simulated abundance data; no study system	Compares standard, design based, buffered, environmentally blocked and spatially blocked partitioning on data generated from a Gaussian process; simulation is the only setting in which the true surface is known, so validation optimism can be measured rather than inferred	Spatial structure and validation findings (3.2)	Verified
Ploton et al. (2020)	Tropical forest biomass, terrestrial	The reference demonstration that random partitioning inflates apparent accuracy under spatial autocorrelation; its 11.8 million observations have no marine counterpart	Validation finding (3.2)	Title
Mi et al. (2017)	Cranes, wetland	One of only two studies quantifying performance below 100 observations; no marine study in the corpus compares algorithms at comparable sample sizes	Coverage and minimum data findings (3.2, 3.4); Tables 2 and 5	Title

Table S3 (continued)

Study	System	Reason for retention	Finding supported	Basis
<i>Sources cited for method provenance or implementation (not corpus candidates)</i>				
Behrens et al. (2018)	Soil, terrestrial	Introduces Euclidean distance fields, the spatial feature construction on which spatial tree ensembles depend	Section 3.4 (coordinate representation)	Journal
Elith et al. (2006)	226 species across six regions, terrestrial	The benchmark comparison for presence only methods; cited for provenance and as the data rich regime this review distinguishes itself from	Introduction	Title
Norberg et al. (2019)	Multiple, primarily non marine datasets	Comparison of 33 methods; cited for the joint model finding that spatial shrinkage variants outperform non spatial joint models	Section 3.4; Tables 2 and 4	Verified
<i>Statistical and machine learning sources with no study system</i>				
Hastie et al. (2009)	Statistical theory	Rotation dependence of axis aligned tree splits	Section 3.4 (coordinate representation)	Verified

Note for reconciliation: main text Section 2.1 states that eight non marine studies were retained under the exception. Table S3 lists five that anchor a finding and three further non marine sources cited for provenance; the count stated in the main text should be made to match the corpus screening records before the preprint is posted.

S1.3 How main text Table 1 was derived

The evidence map (main text Table 1) counts sources cited in support of each contrast, attributed from the citation commands in the main text. A source cited under more than one contrast is counted under each, so the columns do not sum to the number of distinct sources. These counts are a property of the argument as written and can be checked against the printed paper; they are a lower bound on corpus contribution, since a study may inform a finding without being cited where that finding is stated. Marine status follows Table S3.

The counts were derived from the per finding coding of the earlier version of this synthesis (Table S4) and then reassigned to the contrast rows of main text Table 1. Table S4 is retained so that the derivation can be checked.

Table S4. Distinct sources cited in support of each finding in the earlier version of this synthesis, counted from citation commands in that main text.

Finding	Sources cited	Of which marine	Of which crossing both contrasts
Spatial structure before algorithmic complexity	6	3	3
Coverage before sample size	6	4	0
Validation matched to inference target	4	3	0
Decision framework and applications	15	15	0
Method families (Table 2 of the main text)	21	17	1
Quantitative claims (Table 5 of the main text)	9	7	1

The pattern is the content of main text Section 3.1: support for the spatial structure finding rests on six sources, three of them marine, and all three studies crossing spatial treatment against algorithm family are non marine; coverage and validation rest on small numbers of sources reporting the relevant experiment directly; the decision framework draws on fifteen marine sources because it synthesises applied work rather than resting on comparative experiments.

S1.4 Relation to previous reviews

Table S5. Contribution relative to previous reviews of marine species distribution modelling and machine learning in marine systems.

Review	Scope	Quantitative synthesis	Data limitation addressed	Validation guidance	Minimum data thresholds	Evidence map
Robinson et al. (2017)	Marine SDM practice	No	No	Partial	No	No
Melo-Merino et al. (2020)	ML in marine systems	Bibliometric	No	No	No	No
Rubbens et al. (2023)	ML in marine ecology	No	No	Partial	No	No
Pasanisi et al. (2024)	Marine SDM	No	No	No	No	No
Klaassen et al. (2025)	Methodological trends	No	No	No	No	No
This review	Marine SDM under data limitation	Directional, with attributed within study values	Yes, as the organising scope	Yes, matched to inference target	Yes, method specific and graded	Yes

Appendix S2. Extraction protocol, extracted studies and quality assessment

S2.1 Selection for quantitative extraction

Twelve studies were selected on the basis of at least one of four criteria: a direct method comparison on identical data; an explicit assessment of validation optimism; a sample size experiment with associated performance metrics; or an authoritative performance benchmark for a specific method. Selection was made before extraction and was not conditioned on the direction of the reported result.

S2.2 Extraction fields

Sample characteristics. Number of observations, spatial extent, temporal extent, sampling design, response type, prevalence where applicable.

Methodological detail. Algorithms compared, software and version, predictor set, spatial treatment, model complexity settings.

Validation. Partitioning scheme, block size or temporal unit where applicable, whether an independent test set was available, whether the inference target was stated.

Results. Metrics reported, values, direction of the contrast, whether the comparison was within study.

Context. Taxon, ecosystem type, management application if any.

Table S6. Studies selected for detailed quantitative extraction.

II	Study	n	Taxon	System	Comparison	Metrics reported	Result as reported	Quality
	Becker et al. (2020)	4,713	Cetaceans	Pelagic, California Current	BRT against GAM; k fold against leave one year out	Area under the ROC curve; rank correlation	0.85 against 0.83; density surface GAM better on rank correlation; 0.11 partitioning difference	High
	Luan et al. (2020)	40 to 120	Fish	Demersal	Random Forest across sample sizes and sampling designs	Performance stability	Distributed samples outperform clustered samples of equal or greater size	High
	Mi et al. (2017)	33 to 75	Cranes	Wetland	Random Forest against MaxEnt and BRT	Area under the ROC curve	0.82 to 0.89	High
	Thorson et al. (2015)	28 species	Groundfish	Demersal	Geostatistical delta GLMM against design based estimation	Standard error of index	Approximately 20% reduction	Very high
	Thorson (2019a)	About 300 per year	Groundfish	Demersal	Retrospective forecast skill by horizon	Correlation	$r = 0.6$ to 0.7 at one year; 0.3 to 0.4 at five years	High
	Hengl et al. (2018)	Multiple benchmarks	Soil	Terrestrial	Spatial Random Forest against kriging	R^2	Approximately 0.41 against approximately 0.60 on Meuse	High
	Ploton et al. (2020)	11.8 million	Trees	Terrestrial	Random against spatially structured validation	Conceptual and empirical	Ignoring autocorrelation inflates apparent accuracy	Very high
	Mushagalusa et al. (2024)	Simulated	Simulated	Terrestrial	Random against spatial cross validation	Error magnitude	Spatial partitioning yields larger, more honest errors	High
	Grüss et al. (2021)	About 300 per year	Walleye pollock	Demersal, eastern Bering Sea	Spatio temporal model with environmental covariates	Variance explained; correlation	Cold pool approximately 40% ; cross validation $r = 0.72$	High
	Zelli et al. (2025)	1,247	Corals and sponges	Deep sea, New Zealand	BRT with climate projection	Projected habitat retention	Approximately 23% retained under high emissions	High
	Roberts et al. (2017)	Review	Multiple	Multiple	Cross validation framework	Methodological	Basis for partitioning matched to inference target	Very high

Table S6 (continued)

Study	n	Taxon	System	Comparison	Metrics reported	Result as reported	Quality
Flynn and Marshall (2013)	Multiple	Lanternfish	Pelagic	GAM with MaxEnt	Application	Deep pelagic application under sparse sampling	Moderate

S2.3 Quality assessment

Studies were rated across five dimensions: sample adequacy for the inference attempted; validation rigour; reporting clarity, including whether the inference target was stated; statistical robustness; and relevance to data limited marine settings. Very high denotes exemplary performance on all five; high denotes four or five met with minor limitations; moderate denotes three met adequately. Studies rated low were excluded from quantitative extraction and retained, where relevant, for narrative synthesis only.

Quality ratings determined whether a study could anchor an attributed numerical claim in the main text. They were not used as weights, since no pooling or weighted averaging was performed at any stage.

Appendix S3. Metrics and the comparison protocol

S3.1 Metrics encountered in the corpus

Discrimination metrics. The area under the receiver operating characteristic curve measures the probability that a randomly chosen presence receives a higher predicted score than a randomly chosen absence, bounded between 0.5 and 1. Its dominance reflects three convenient properties: independence from any threshold choice, boundedness across study contexts, and near universal implementation. Three limitations bear on synthesis. It is insensitive to calibration, so a model can discriminate well while producing biased probabilities. It depends on the spatial extent from which absences or background points are drawn, so values are not comparable across studies with different domains. It weights omission and commission errors equally, which rarely matches a conservation decision context (Lobo et al., 2008; Guillera-Arroita et al., 2015). The true skill statistic partially addresses prevalence dependence (Allouche et al., 2006).

Error metrics. Root mean square error and mean absolute error are scale dependent and not comparable across systems without normalisation. R^2 is bounded but depends on the variance structure of the data and on whether it was computed in training or in validation.

Management relevant metrics. For abundance and density applications, agreement between a model derived index and an independent estimator, expressed as a correlation or a relative standard error, is often more decision relevant than any pointwise predictive metric.

S3.2 Why no pooled estimate is reported

No defined transformation relates a difference in a discrimination metric to a difference in an error metric. Any mapping between them requires assumptions about prevalence, the shape of the predicted score distribution and the loss function, none of which is recoverable from a published summary. Averaging across studies that report different metric families therefore produces a quantity with no interpretation, and averaging within a metric family across studies with different taxa, prevalences, domains and validation designs produces a quantity whose reference population is undefined.

An earlier version of this synthesis reported cross study ranges for the spatial structure effect and the algorithm choice effect, expressed in the area under the curve. Those ranges were assembled by treating values reported in different metric families as approximately commensurate and averaging across them with weights reflecting study size and quality. That procedure has no defensible basis, and the resulting values have been withdrawn rather than reframed. Table S7 records each claim examined, the sources previously cited in its support, what those sources actually report, and the verdict.

S3.3 The protocol applied

1. Within study contrasts are reported with attribution to the source, its taxon, its system and the metric that source used.
2. No value is averaged, weighted or otherwise combined across studies.
3. Where studies agree in the direction of a contrast but not in metric or magnitude, the direction is reported together with the number of studies supporting it, and no magnitude is attached.
4. Every numerical value in the main text appears in main text Table 5 with its source and its scope of generalisation.
5. Ranges, where they appear, describe the values reported by named studies and are not confidence intervals.

This protocol is a response to the current state of the evidence base. It is not offered as a general method, and it does not solve the underlying problem, which requires standardised multi metric reporting in primary studies (Appendix S7).

Appendix S4. Survey design, effort and data quality; environment specific scaling; two stage assessment workflows

S4.1 Survey design: coverage, effort and data quality

This appendix gives the extended treatment of the material summarised in main text Section 3.4.

Coverage against total sample size. Spatially dispersed observations spanning the full environmental gradient outperform concentrated samples of equal or larger size (Robinson et al., 2011; Luan et al., 2020). The mechanism is representation of environmental space: clustered samples leave regions unobserved, so prediction there is extrapolation rather than interpolation, and the model is being asked a question its training data cannot answer regardless of how many records that training data contains.

Stratification. Environmental stratification allocates effort proportionally across the gradients structuring the response, typically sea surface temperature, depth zone and productivity regime. Geographic stratification prioritises breadth across regions and is the practical approximation where the structuring gradients are not known in advance. Temporal stratification samples across seasons and years, capturing phenological variation and supplying the replication that spatio temporal methods require structurally. Under budget constraint, breadth of coverage should be preferred to intensity of local effort (Canonico et al., 2019).

Effort and detection. Sample quality interacts with quantity. Geostatistical models represent varying effort through the likelihood; tree based methods treat effort as uniform unless it is supplied as a predictor, so an unmodelled effort gradient is absorbed into the fitted habitat relationship. Imperfect detection reduces effective sample size relative to the nominal count, and the reduction is severe where detection probability is low. Taxonomic certainty and covariate matching bear on effective sample size in the same way.

Priorities under constraint. In the order in which they bind: broad spatial coverage ensuring environmental space is represented; sample size meeting the method specific minimum; temporal replication where a spatio temporal method is intended; and environmental covariates at appropriate resolution and temporal alignment.

Below the threshold. With fewer than roughly 30 observations, practitioners should either accept an exploratory analysis with transparent limitations, or defer formal modelling (Guillera-Arroita et al., 2015). Presence only methods can provide initial insight framed as hypothesis generating rather than hypothesis testing. Expert derived habitat suitability assessment, built from literature and local knowledge, is a legitimate alternative where data are insufficient. Where modelling proceeds, inference should be restricted to coarse scale pattern with uncertainty stated.

S4.2 Environment specific scaling of minimum sample size

Main text Table 2 gives minimum data requirements by method with environment specific scaling merged in. The scaling factors are author estimates informed by patterns in the reviewed literature. They are not empirically derived multipliers from controlled experiments, no study in the corpus reports them, and they should be read as an ordering of environments by expected data requirement rather than as coefficients. They are recorded as declared estimates in Appendix S4.5.

Table S7. Environment specific scaling of minimum sample size requirements. Multipliers are author estimates, not empirical coefficients (grade E).

Environment	Scaling factor	Random Forest minimum	Geostatistical minimum	Rationale
Pelagic	About 1.5 times	About 75 observations	About 45 stations	High spatio temporal variability from currents, mesoscale eddies and frontal dynamics; mobile species with large ranges; weaker habitat associations
Demersal and benthic	1.0 (reference)	About 50 observations	About 30 stations	Stronger habitat structure and more predictable species substrate associations
Coastal	About 0.75 times	About 40 observations	About 25 stations	Accessible sampling, strong environmental gradients, greater data availability
Deep sea	0.6 to 1.0 times	30 to 50 observations	Not recommended	Extreme data limitation; prefer presence absence over abundance, and methods that function at minimal sample sizes

The ordering rests on the variability of the environment rather than on accessibility alone. Pelagic systems weaken habitat associations and inflate residual variability (Bell et al., 2014; Robinson et al., 2011); demersal and benthic systems supply the reference case; coastal work benefits from accessibility and steep gradients (Canonico et al., 2019). Deep sea applications face the severest limitation, and the appropriate response is to narrow the inferential target, predicting occurrence rather than abundance and coarse pattern rather than fine surfaces, rather than to relax the performance standard a model is held to.

S4.3 Index standardisation and two stage assessment workflows

One class of application warrants separate treatment, because the appropriate method choice is determined by a downstream consumer rather than by the prediction task itself. In two stage assessment workflows a model is fitted to survey data to produce a time series of annual abundance

indices, and those indices are then passed to a separate assessment model that combines them with catch data, biological information and population dynamics assumptions (Thorson, 2019b). The structural assumptions of the downstream consumer constrain the upstream choice.

The binding constraint is the temporal error structure of the indices. Assessment models generally treat the index as a series of observations with independent year specific errors, and their likelihood, goodness of fit diagnostics, retrospective analysis and uncertainty propagation all rest on that independence. A model that imposes strong temporal autoregressive structure on the abundance field produces indices that are correlated across years, which are not plausible draws from independent year specific errors. The assessment then understates true uncertainty, because correlated index errors appear more informative than they are, and correlation propagates into estimated trends as retrospective pattern.

This is the reason for a specific structural choice in the geostatistical packages in main text Table 2 (Thorson, 2019b). They permit rich spatial structure through Gaussian Markov random fields on the spatial random effects, which is what makes them useful, while not imposing strong temporal autoregression on the year to year evolution of the abundance field. The annual indices that result are spatially informed and temporally independent in their error structure. Methods that impose explicit AR(1) or random walk structure on the abundance field yield smoother indices that look more attractive in isolation and are less appropriate for this workflow.

Two qualifications. The recommendation applies to two stage workflows only. Where the model is itself the final product, in habitat mapping, protected area design, dynamic management or short term forecasting, temporal autoregressive structure may improve prediction at the cost of an independence no downstream consumer requires. And the recommendation concerns temporal structure, not spatial: spatial structure should be included throughout. Practitioners should establish whether model output will feed an assessment before committing to a method's structural choices.

Appendix S5. Computational cost and software

S5.1 Computational cost

Computational cost bears on method selection wherever prediction must be updated frequently, as in dynamic management (main text Table 4). Table S9 gives approximate requirements. Scaling behaviour matters more than absolute timings, since the latter depend on hardware.

Table S8. Approximate computational requirements. Timings assume typical marine datasets of 100 to 10,000 observations with 5 to 15 predictors, and depend on model complexity, spatial resolution and hardware.

Method	Fitting time	Prediction time per surface	Scaling behaviour
GLM	Seconds	Seconds	Linear
GAM (mgcv)	Seconds to minutes	Seconds	Sub linear with bam
Random Forest (ranger)	Seconds to minutes	Seconds	Linear in observations and predictors
BRT (gbm)	Minutes	Seconds	Linear in tree number
VAST	Minutes to hours	Minutes	Super linear in knots
sdmTMB	Minutes to about 30 minutes	Seconds to minutes	Super linear in mesh complexity
tinyVAST	Minutes to hours	Seconds to minutes	Super linear in mesh complexity and species number
BART	Minutes	Seconds	Linear in tree number
Hmsc	Hours; MCMC with 2 to 4 chains and 1,000 or more posterior samples	Minutes for posterior predictive draws across the prediction grid	Super linear in species richness and site count
sjSDM	Minutes to about 30 minutes; variational inference	Seconds to minutes	Linear to sub linear in species count; GPU accelerable

The operational consequence is the one drawn in main text Table 4. Tree based methods dominate near real time systems because prediction is fast and requires no mesh, not because they predict better. Where calibrated uncertainty is required in such a system, the cost of a geostatistical alternative falls on fitting rather than prediction, which is why separating frequent field updates from infrequent hyperparameter re estimation is the plausible route to operational deployment.

S5.2 Software and implementation

Table S9. Software implementations for the method families in main text Table 2, with the functions that carry the spatial component and notes specific to marine application. Package versions change; access dates are September 2026.

Method	R packages	Key functions and workflow	Python	Notes for marine applications
Random Forest	ranger; randomForest	ranger() for fast fitting with variable importance; randomForest() as the reference implementation; spatial predictors are constructed by the analyst	scikit-learn	ranger preferred above roughly 10,000 observations for its C++ backend; use permutation importance where predictors are correlated, which is usual with marine covariates
Spatial Random Forest	spatialRF; RFsp via landmap; waywiser; RandomForestsGLS	rf_spatial() fits with spatial predictors; rf_evaluate() for spatial evaluation; RandomForestsGLS provides the dependent data formulation	verstack	Distance predictors must be computed against an ecologically meaningful reference, not the coordinate grid alone; distance to shore, shelf break or depth contour are the usual choices
VAST	VAST; FishStatsUtils; TMB (Kristensen et al., 2016)	fit_model() is the primary interface; make_data() and make_model() for configuration	limited	Knot number controls both the resolution of the spatial field and the fitting cost; start coarse and refine, since super linear scaling makes over specification expensive
sdmTMB	sdmTMB (Anderson et al., 2025); TMB; fmesher	sdmTMB() uses a formula interface comparable to glm or mgcv; make_mesh() constructs the SPDE mesh; sanity() reports convergence checks	limited	The recommended entry point for practitioners moving from mgcv; delta families handle the zero inflation typical of trawl and transect data; get_index() produces standardised abundance indices
tinyVAST	tinyVAST (Thorson et al., 2025); TMB; fmesher	tinyVAST() extends the sdmTMB conventions to multivariate and structural equation specifications	limited	For multi species work without VAST's full specification overhead; cross species spatial fields borrow strength toward sparsely observed species
Spatial GAMs	mgcv (Wood, 2017); dsm (Miller et al., 2022); gratia	gam() and bam() with s(x, y, bs = "tp") or te(x, y); dsm for density surface models	pyGAM	dsm is the standard route for line transect survey data, which covers most cetacean and seabird applications
MaxEnt	maxnet; ENMeval; dismo	maxnet() is the native R implementation requiring no Java; ENMevaluate() automates tuning and background selection	elapid	Background extent defines the domain against which discrimination is measured and must match the area over which prediction is required; it drives apparent performance more than any tuning setting
Boosted Regression Trees	gbm; dismo; xgboost; lightgbm; caret	gbm.step() performs cross validated selection of tree number with ecologically appropriate defaults (Elith et al., 2008)	xgboost, lightgbm	gbm.step() applies ecologically appropriate defaults and cross validated tree selection; prefer it to direct fitting for distribution work

Table S10 (continued)

Method	R packages	Key functions and workflow	Python	Notes for marine applications
BART	embarcadero; dbarts	bart.step() for variable selection in a distribution modelling context; bart() and bart2() for fitting	bartpy	Posterior uncertainty without MCMC cost at community scale
Graphical interface	GLOSSA (Mestre-Tomás et al., 2025)	Shiny application providing a no code interface to BART based models with spatial blocking	none	Useful where analysis must be handed to non specialist staff, or where a workflow is demonstrated to stakeholders
Joint SDM, hierarchical	Hmsc	Hmsc() with spatial random effects; createStudyDesign() takes spatial coordinates; computeAssociations() returns residual species correlations; MCMC sampling, hours for large communities	limited	Benefit concentrates where a rare target species co occurs with better sampled congeners; see Table S11
Joint SDM, variational	sjSDM (Pichler and Hartig, 2021)	sjSDM() performs fast variational inference through a PyTorch backend	via the same backend	Faster than full MCMC for large communities, at some cost in posterior accuracy; see Table S11
Joint SDM, ordination	boral	boral() for Bayesian latent variable ordination of multivariate abundance data	limited	Ordination and community description rather than spatial prediction; see Table S11
Ensembles	biomod2; caretEnsemble; stacks	BIOMOD_Modeling() and BIOMOD_EnsembleModeling() for multi model platforms	scikit-learn	Appropriate where candidates are inseparable under matched validation (main text 3.4), not as a default
Spatial cross validation	blockCV (Valavi et al., 2019); spatialsample; ENMeval	cv_spatial() performs block partitioning with automated range estimation; cv_spatial_autocor() estimates the range; cv_cluster() supplies spatial clustering where blocks cannot be formed; block size should exceed the residual autocorrelation range	verstack, spacy	Block size must exceed the residual autocorrelation range, which should be estimated rather than assumed; where the subsample extent is smaller than the range, blocking at the recommended size is impossible and this should be reported
Spatial diagnostics	spdep; gstat; DHARMA	moran.test() and moran.mc() for Moran's I on residuals; variogram() for empirical variograms; residual simulation for distributional checks	pysal	Moran's I on residuals rather than on the raw response; residual structure is what bears on the model

Table S10 (continued)

Method	R packages	Key functions and workflow	Python	Notes for marine applications
Graph neural networks	torch	Limited R support; most implementations use Python	PyTorch Geometric	Noted for completeness; data requirements exceed the settings this review addresses

Table S10. Joint species distribution model frameworks, distinguished by where each applies shrinkage and how inference is performed. The distinction matters under data limitation, because shrinkage on environmental responses is the mechanism by which sparsely observed species borrow strength from better sampled ones.

Framework	R package	Shrinkage on environmental responses	Shrinkage on spatial latent variables	Inference	Recommended marine use
Hierarchical, with spatial random effects	Hmsc	Yes; improves estimation for species with sparse observations	Yes; captures residual spatial autocorrelation across the community	Full MCMC; hours	Demersal fish and benthic assemblages; community level conservation planning where a full posterior is required. Norberg et al. (2019) report the spatial variant outperforming non spatial joint models
Hierarchical, without spatial random effects	Hmsc	Yes	No	Full MCMC	Only where residual spatial autocorrelation is genuinely weak, or where computation precludes spatial latent factors; otherwise prefer the spatial variant
Variational	sjSDM (Pichler and Hartig, 2021)	Yes, through regularisation	Yes, through spatial latent variables	Variational inference; minutes	Large communities above roughly 30 species; iterative assessment against tight timelines; faster than full MCMC at some cost in posterior accuracy
Bayesian ordination	boral	Yes, through latent variables	No spatial structure in the base implementation	Full MCMC	Compositional analysis and ordination; community description rather than spatial prediction
Multi species geostatistical	VAST (Thorson, 2019b)	Implicit, through shared random effects	Yes; vector autoregressive structure on spatial fields across species	Empirical Bayes by Laplace approximation; minutes to hours	Multi species survey data from structured programmes, and where output feeds an assessment model (Appendix S5.3)
Multi species geostatistical, formula interface	tinyVAST (Thorson et al., 2025)	Implicit, through shared random effects	Yes; cross species spatial fields with optional simultaneous effects	Empirical Bayes; minutes to hours	For practitioners moving from sdmTMB to multi species work without the full specification overhead

Appendix S6. Reporting protocol

We recommend that primary studies contributing to future syntheses report, at minimum: the inference target the validation was designed to estimate; the partitioning scheme and, for spatial blocking, the block size and its relation to the estimated autocorrelation range, including whether the study extent permitted blocking at that size; at least one discrimination metric and one error or calibration metric where the response type permits both; the spatial and environmental coverage of training data relative to the prediction domain; and the proportion of the prediction domain requiring extrapolation. The ODMAP protocol (Zurell et al., 2020) provides the structure for the first four; the metric convention is the missing element.

Supplementary references

References cited only in this Supporting Information; all others are in the main text reference list.

Allouche, O., Tsoar, A., & Kadmon, R. (2006). Assessing the accuracy of species distribution models: Prevalence, kappa and the true skill statistic (TSS). *Journal of Applied Ecology*, 43(6), 1223–1232.

Bahn, V., & McGill, B. J. (2013). Testing the predictive performance of distribution models. *Oikos*, 122(3), 321–331.

Behrens, T., Schmidt, K., Viscarra Rossel, R. A., Gries, P., Scholten, T., & MacMillan, R. A. (2018). Spatial modelling with Euclidean distance fields and machine learning. *European Journal of Soil Science*, 69(5), 757–770.

Bell, R. J., Richardson, D. E., Hare, J. A., et al. (2014). Disentangling the effects of climate, abundance, and size on the distribution of marine fish: An example based on four stocks from the Northeast US shelf. *ICES Journal of Marine Science*, 72(5), 1311–1322.

Flynn, A. J., & Marshall, N. J. (2013). Lanternfish (Myctophidae) zoogeography off eastern Australia: A comparison with physicochemical biogeography. *PLOS ONE*, 8(12), e80950.

Kristensen, K., Nielsen, A., Berg, C. W., et al. (2016). TMB: Automatic differentiation and Laplace approximation. *Journal of Statistical Software*, 70(5), 1–21.

Ovaskainen, O., & Abrego, N. (2020). *Joint Species Distribution Modelling: With Applications in R*. Cambridge University Press.

Pichler, M., & Hartig, F. (2021). A new joint species distribution model for faster and more accurate inference of species associations from big community data. *Methods in Ecology and Evolution*, 12(11), 2159–2173.

Redfern, J. V., Ferguson, M. C., Becker, E. A., et al. (2006). Techniques for cetacean habitat modeling. *Marine Ecology Progress Series*, 310, 271–295.

- Rooper, C. N., Sigler, M. F., Goddard, P., et al. (2016). Validation and improvement of species distribution models for structure-forming invertebrates in the eastern Bering Sea with an independent survey. *Marine Ecology Progress Series*, 551, 117–130.
- Rooper, C. N., Wilborn, R., Goddard, P., et al. (2018). Validation of deep-sea coral and sponge distribution models in the Aleutian Islands, Alaska. *ICES Journal of Marine Science*, 75(1), 199–209.
- Rooper, C. N., Williams, K., Towler, R., et al. (2025). Lessons learned from testing the predictions of species distribution models for deep-sea corals and sponges in the Gulf of Alaska with comparisons to other Alaska ecosystems. *ICES Journal of Marine Science*, 82(10).
- Zurell, D., Franklin, J., König, C., et al. (2020). A standard protocol for reporting species distribution models. *Ecography*, 43(9), 1261–1277.